

Auto-Failover Website Monitoring System On AWS

Detect failure → Send alert → Redirect traffic

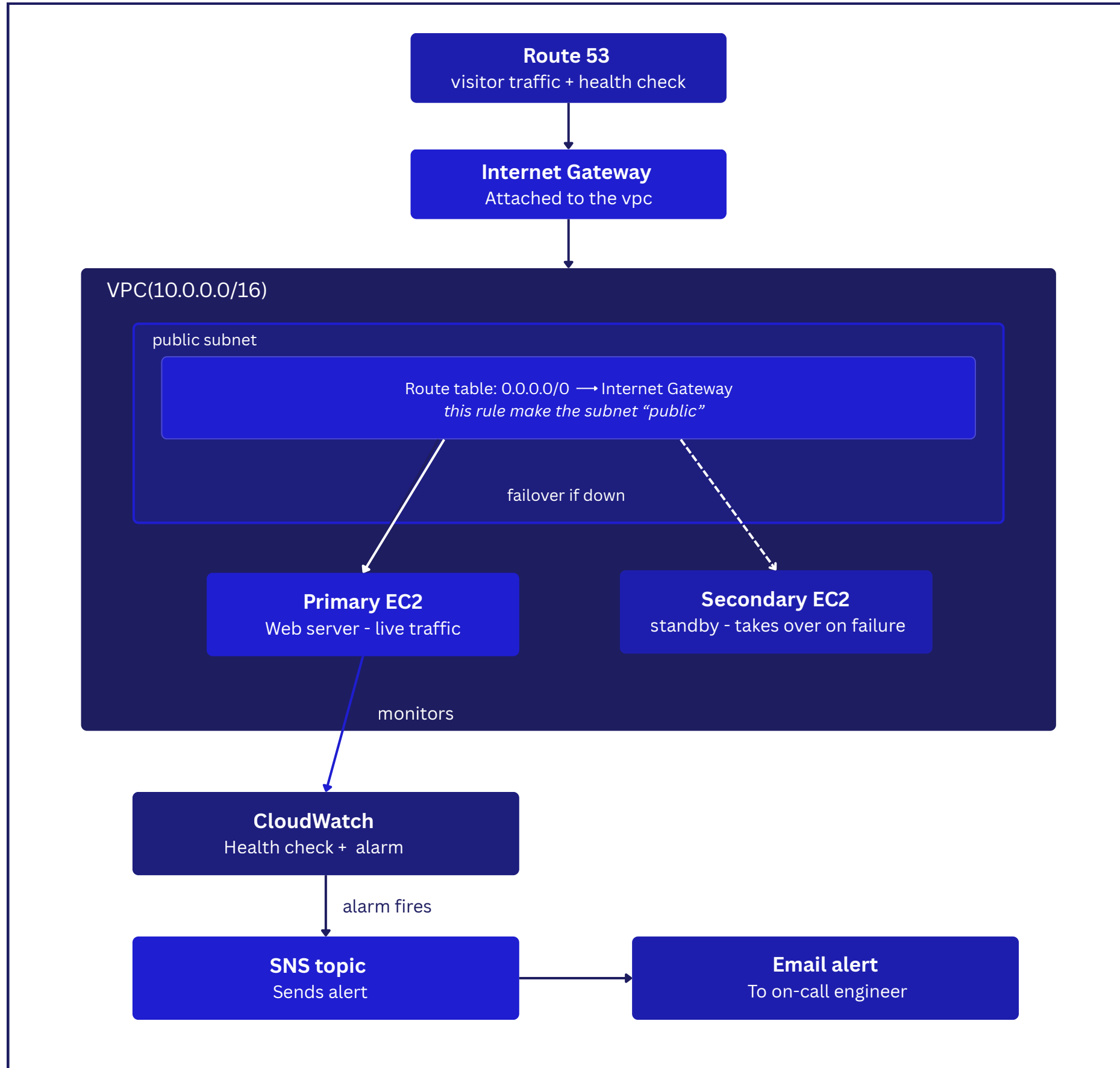
VPC • EC2 • Route 53 • CloudWatch • SNS

Overview and Services Used

A Route 53 health check watches a primary web server. When it stops responding, a CloudWatch alarm fires and SNS emails the engineer. A standby server is ready to take over using Route 53 failover routing.

Service	Purpose
VPC	10.0.0.0/16 network, public subnet, Internet Gateway, route table
EC2 (x2)	Primary and secondary Apache web servers (Amazon Linux 2023)
Elastic IP	Fixed public IP for each server
Route 53	HTTP health check on the primary; failover routing design
CloudWatch	Alarm on the HealthCheckStatus metric (us-east-1)
SNS	Email notification when the alarm triggers

System Architecture



Traffic flow

- Visitors reach the site through Route 53 and the Internet Gateway.
- Traffic goes to the primary EC2; the secondary is on standby for failover.

Alert flow

- The Route 53 health check monitors the primary server.
- CloudWatch alarm fires when the health check fails.
- SNS sends an email alert to the on-call engineer.

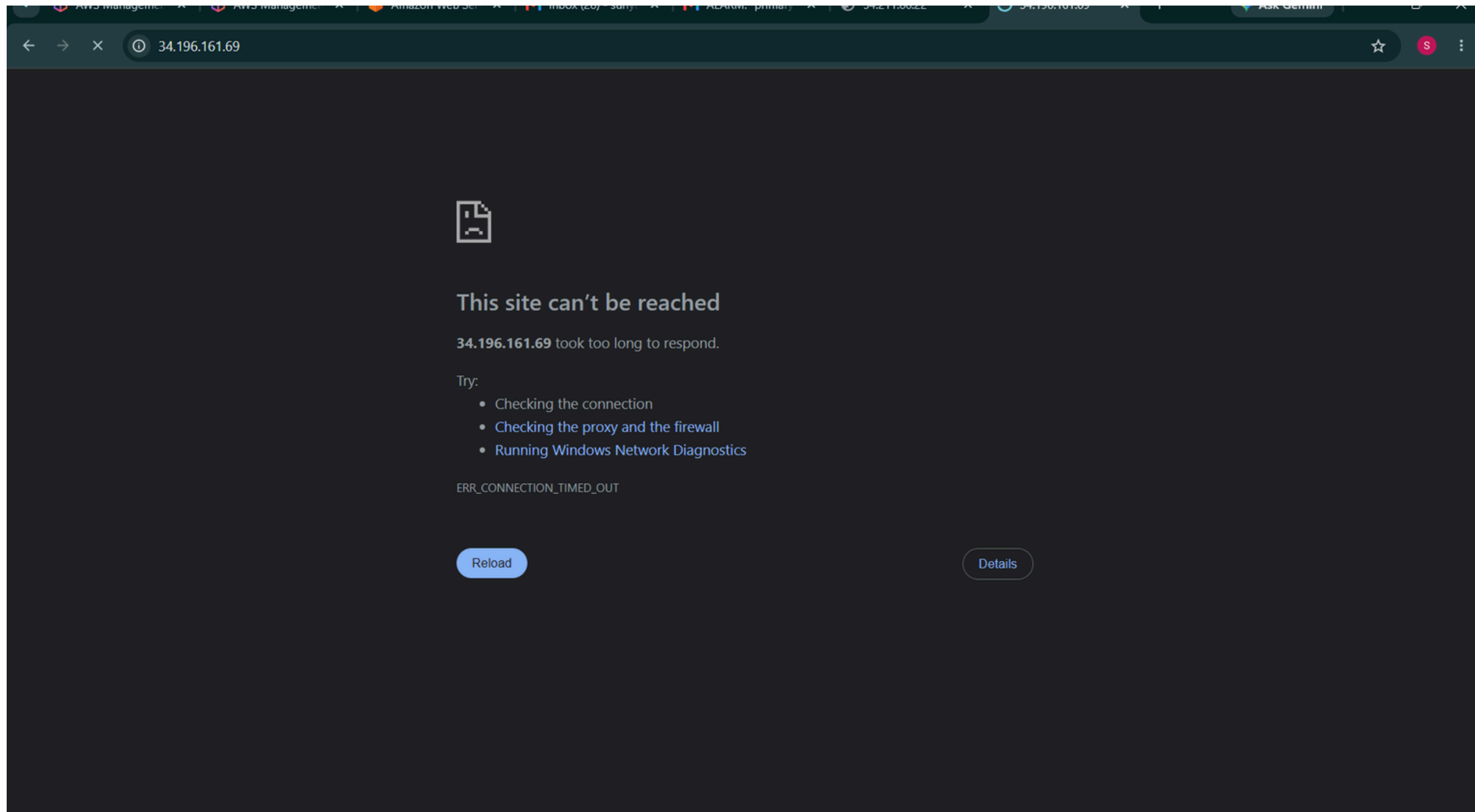
Implementation Steps

- Created a VPC with a public subnet, Internet Gateway and a 0.0.0.0/0 route to the gateway.
- Created a security group: HTTP (80) from anywhere, SSH (22) from my IP.
- Launched two EC2 instances with Apache installed via user data, each with an Elastic IP.
- Created an SNS topic with a confirmed email subscription.
- Created a Route 53 health check (HTTP, port 80) on the primary server
- Created a CloudWatch alarm: HealthCheckStatus lower than 1 for 1 minute, notify SNS.
- Designed the Route 53 failover records: primary (with health check) and secondary.

Failure test: the primary EC2 instance was stopped to simulate an outage.

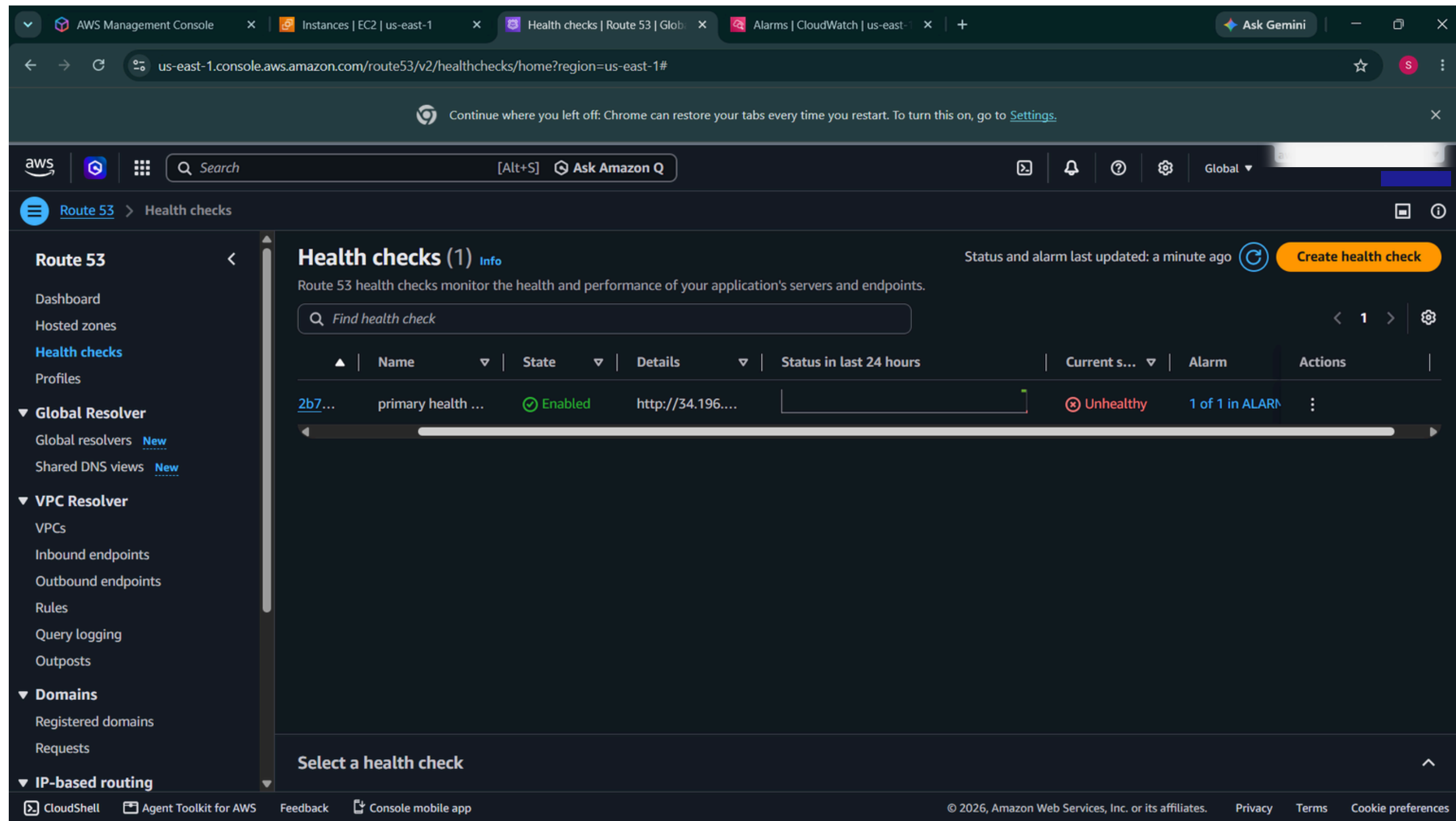
Note: detection and alerting were tested live; the DNS failover records were designed but not deployed, because that requires a registered domain

1. Primary server unreachable



After stopping the instance, the primary Elastic IP stops responding (ERR_CONNECTION_TIMED_OUT).

2. Route 53 health check : Unhealthy

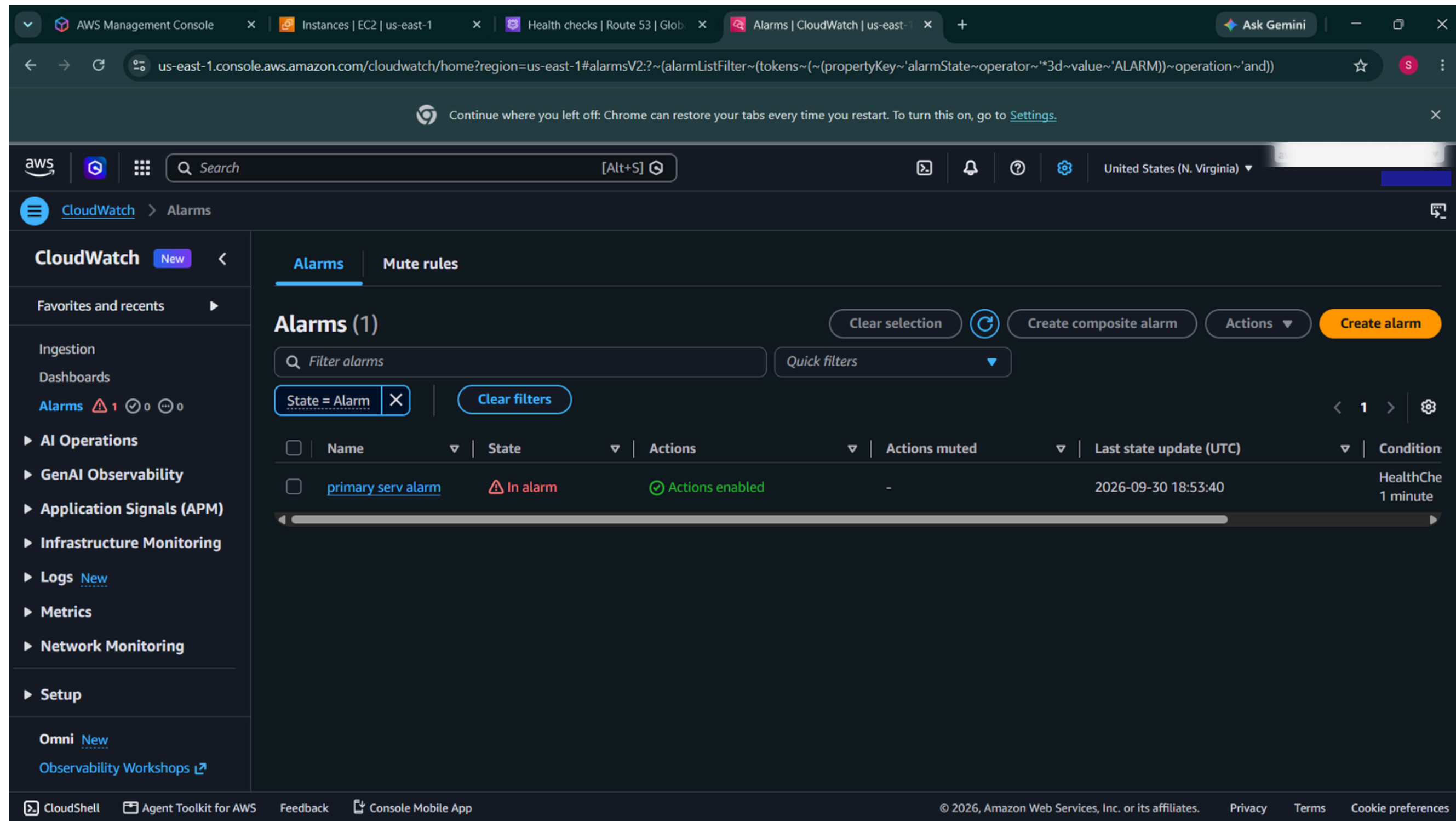


The screenshot displays the AWS Management Console interface for Route 53 health checks. The left-hand navigation pane lists various services, with 'Route 53' selected. Under 'Route 53', the 'Health checks' option is highlighted. The main content area shows 'Health checks (1)' with a status of 'Enabled'. A table lists the health check details, including its name, state, details, status in the last 24 hours, current status, and alarm. The current status is 'Unhealthy', and the alarm is '1 of 1 in ALARM' state. A 'Create health check' button is visible in the top right corner.

Name	State	Details	Status in last 24 hours	Current s...	Alarm	Actions
primary health ...	Enabled	http://34.196...		Unhealthy	1 of 1 in ALARM	

The health check detects the failure and shows 1 of 1 alarm in ALARM state.

3. CloudWatch alarm : in alarm



The screenshot shows the AWS Management Console for the CloudWatch Alarms page in the us-east-1 region. The left sidebar contains navigation links for CloudWatch, Alarms, and various monitoring tools. The main content area displays a list of alarms, with one alarm, 'primary serv alarm', currently in the 'In alarm' state. The alarm's condition is 'HealthCheck 1 minute'.

CloudWatch Alarms (1)

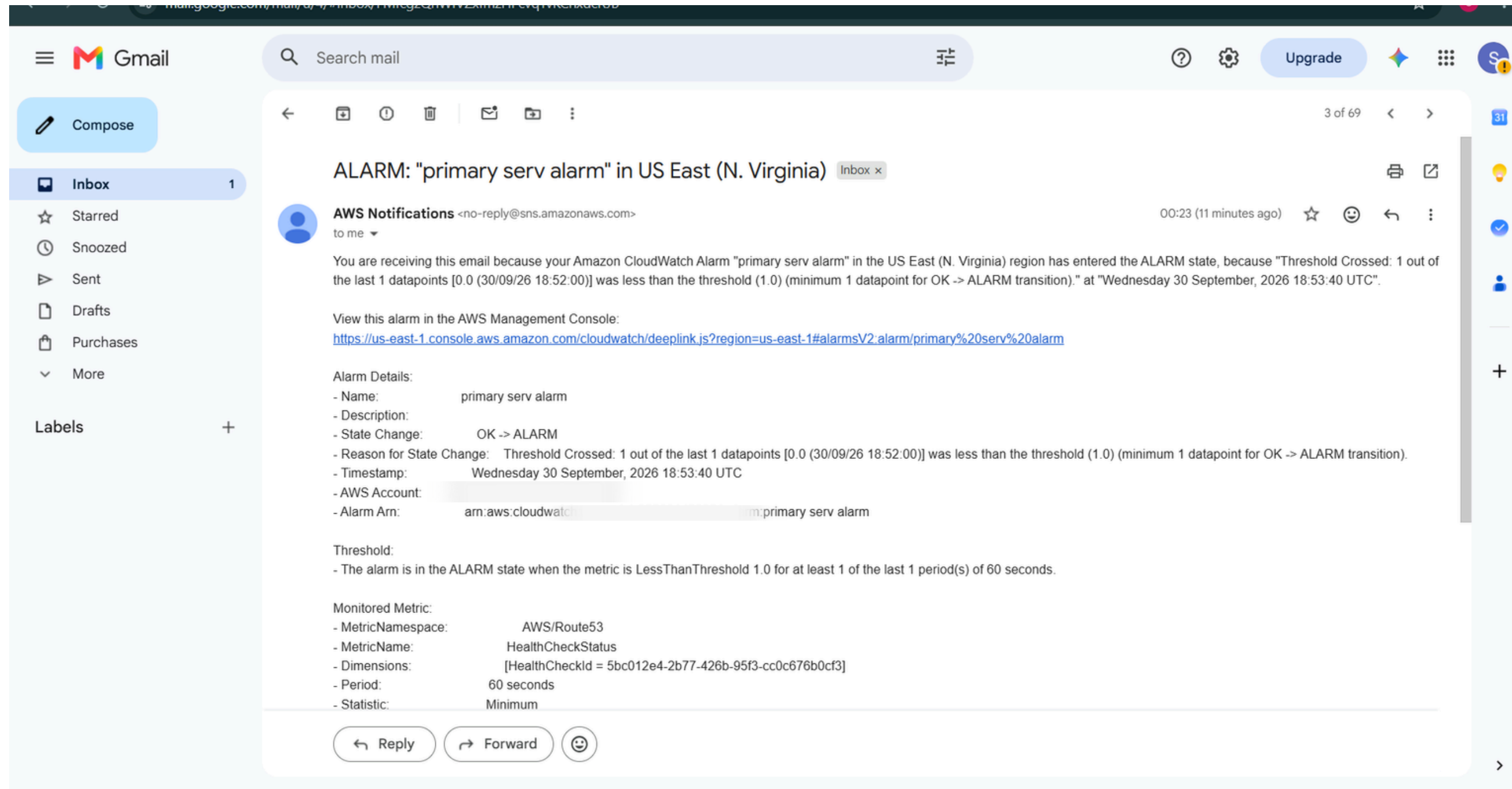
Filter alarms: Quick filters:

State = Alarm

☐	Name	State	Actions	Actions muted	Last state update (UTC)	Condition
☐	primary serv alarm	In alarm	Actions enabled	-	2026-09-30 18:53:40	HealthCheck 1 minute

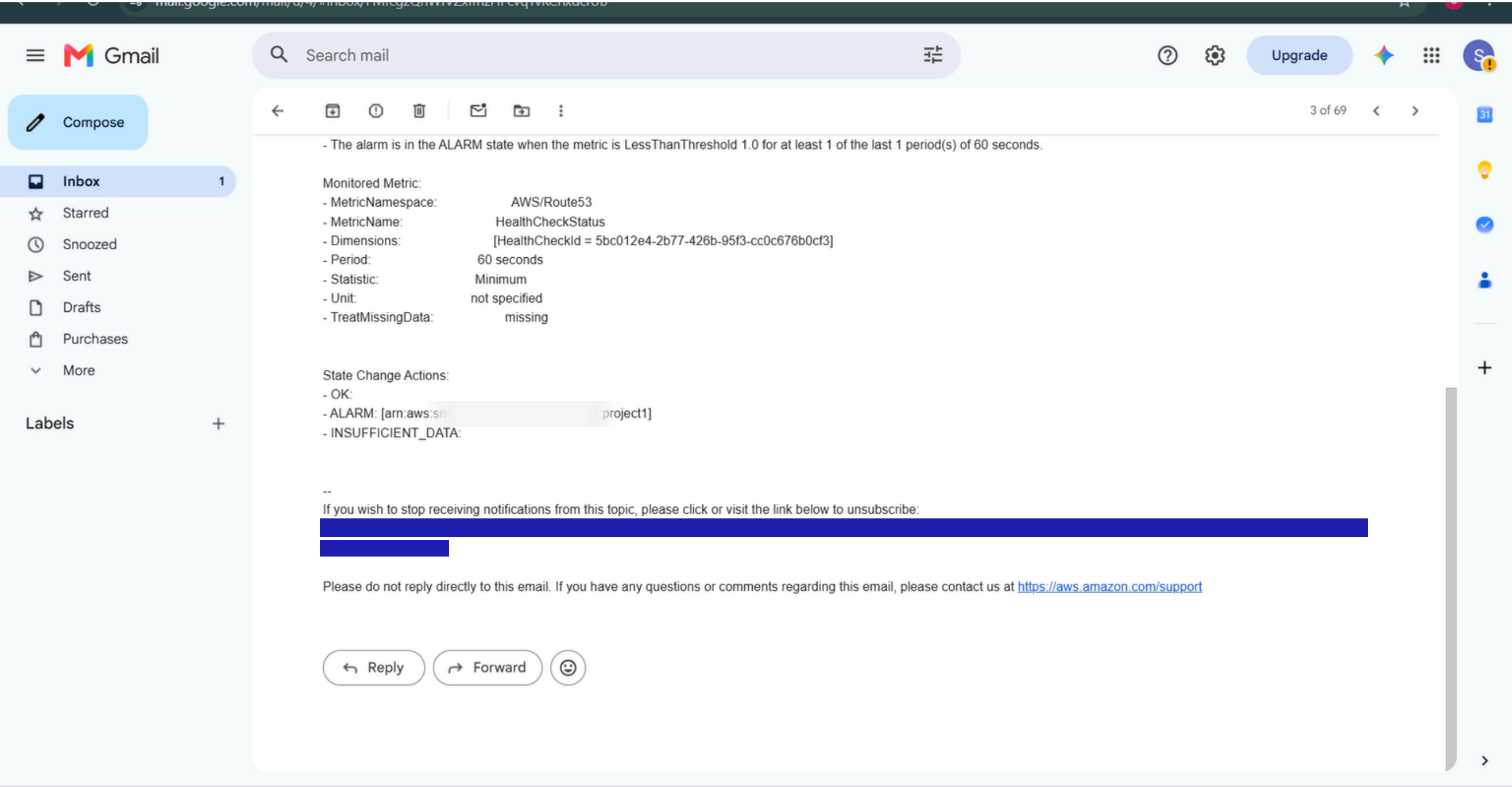
The alarm on the HealthCheckStatus metric moved to In alarm in us-east-1.

4. SNS email alert



AWS Notifications emailed the OK to ALARM state change.

5. Alert email details



Metric AWS/Route53 HealthCheckStatus, statistic Minimum, period 60 seconds, action sent to the SNS topic

6. Secondary server running



The standby server responds on its own Elastic IP and is ready to take over.

What I Learned and Next steps

What I learned

- How health checks, alarms and notifications connect across AWS services.
- Route 53 health check metrics live in us-east-1.
- Public subnet networking: Internet Gateway, route tables, security groups.
- Cost control: deleted all resources after testing.

Future improvements

- Deploy failover records with a registered domain.
- Spread servers across multiple Availability Zones.
- Add a load balancer and Auto Scaling.
- Provision with Terraform or CloudFormation.